

change in flux is negative.

9(a)(i)	Two pieces of evidence: 1. For a given metal, there is a threshold frequency. If the frequency of the incident electromagnetic radiation is below this threshold frequency, no emission can occur, no matter how intense is the electromagnetic radiation. 2. There is no time lag between the arrival of the incident photons and the emission of photoelectrons, for photons of energy above the work function of the metal.
9(a)(ii)	The lines in the emission or absorption spectra of isolated atoms corresponds only to specific wavelengths of light. Since light is composed of photons, this means that only specific energies are emitted or absorbed by the atoms, which suggests that there are fixed energy transitions between discrete energy levels responsible for the production or absorption of such photons.
9(b)(i)	Energy of a photon of purple light $E = \frac{hc}{\lambda}$ $= \frac{(6.67 \times 10^{-34})(3.00 \times 10^8)}{340 \times 10^{-9}}$ $= 5.885 \times 10^{-19} \text{ J}$ $= 3.678$ $\approx 3.68 \text{ eV}$

No.	Solution
9(b)(ii)	The next highest electron energy level in the atom is at -3.4 eV. For an electron at this energy level to de-excite to the -13.6 eV energy level, the electron will lose 10.2 eV of energy which is 2.8 times larger compared to the answer in (b)(i) corresponding to the shortest wavelength in the visible spectrum. Hence, a visible line spectrum does not result from electrons de-exciting to the -13.6 eV energy level.
9(c)(i)	Probability of decay in a time of 1.0 day $\lambda = \frac{\ln 2}{t_{\frac{1}{2}}} = \frac{\ln 2}{53}$ $= 0.01308$ ≈ 0.0131
9(c)(ii)	Initial number of undecayed beryllium-7 in the radioactive sample, $N_0 = \frac{5.7 \times 10^{-12}}{7 \times 1.66 \times 10^{-27}}$ $= 4.905 \times 10^{14}$ Use $N = N_0 \exp(-\lambda t)$, number of undecayed beryllium-7 nuclei in sample after 120 days, $N = (4.905 \times 10^{14}) \exp(-0.0131 \times 120)$ $= 1.018 \times 10^{14}$ $\approx 1.02 \times 10^{14}$
9(c)(iii)	The beryllium-7 nuclei decays through electron capture whereby the captured electrons comes from the K-shell because those electrons are closest to the nucleus. The decay process produces an emission of a γ-ray photon. Since the number of nucleons in the beryllium-7 nucleus does not vary after the decay, the number of beryllium-7 nuclei in the sample does not change.
9(d)	The mass of the decayed beryllium-7 nucleus remains unchanged after decay. Momentum of γ-ray , $p_\gamma = \frac{h}{\lambda} = \frac{h}{\frac{hc}{E_\gamma}}$ $= \frac{E_\gamma}{c}$ $= \frac{0.48 \times 10^6 \times 1.60 \times 10^{-19}}{3.00 \times 10^8}$ $= 2.56 \times 10^{-22} \text{ kg m s}^{-1}$

No.	Solution
	$p_{\text{Be}} = p_{\gamma} \Rightarrow m_{\text{Be}} v_{\text{Be}} = p_{\gamma}$ $v_{\text{Be}} = \frac{p_{\gamma}}{m_{\text{Be}}}$ $= \frac{2.56 \times 10^{-22}}{7 \times 1.66 \times 10^{-27}}$ $= 2.20 \times 10^3 \text{ m s}^{-1}$